Project Planning & Management

PROJECT OVERVIEW

PROJECT TITLE:

Delivery Savaari (auto obstacle detecting delivery car)

TEAM MEMBERS:

Devoshree, Nirzara, Vruddhi

BRIEF DESCRIPTION:

Delivery Savaari is a auto obstacle detecting delivery car with an ultra sound sensor that detects obstacles, its designed to transport items from one location to another. Inspired by Chinese self-driving delivery robots and reimagined in an Indian context, the car is styled as a traditional Indian truck (lorry) complete with iconic traditional patterns on the side panels.
The car supports three operating modes: Manual Control via a custom Bluetooth app, Auto Obstacle Avoidance using an HC-SR04 ultrasonic sensor, and Predefined Path execution for fixed delivery routes. Electronics components are placed in a laser cut MDF enclosure, with a separate transparent acrylic delivery box mounted on the top. A physical switch on the exterior controls main battery power and basically initializes the whole system.

TEAM ROLES & RESPONSIBILITIES

AS DISCUSSED

MEMBER NAME	ROLE	RESPONSIBILITIES
Nirzara	Designer, Programmer & Engineer	Predefined Path Storage, Light and sound indicators, App code and design, Laser cutting the circuit/delivery box parts
Devoshree	Designer, Programmer & Engineer	Motor Control Functions, Obstacle Avoidance Logic, Path Execution Function, Main Loop, Circuit case & box modelling, Assembling final structure
Vruddhi	Designer, Programmer & Engineer	Imports & Pin Setup, Initialize Bluetooth (BLE Server), Ultrasonic Sensor Function

* The responsibilities listed above reflect the initial distribution at the start of the project. In practice, the division of work shifted as the project progressed. This section documents what was actually completed by me.

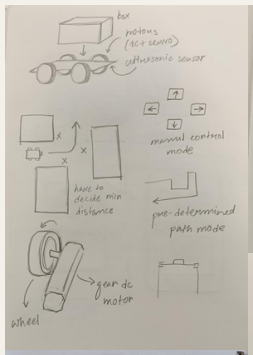
MY ACTUAL CONTRIBUTION

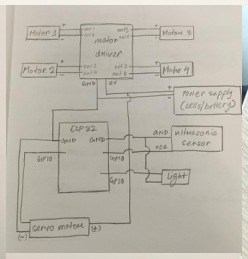
- Handled the majority of the technical work across software, hardware, and fabrication.

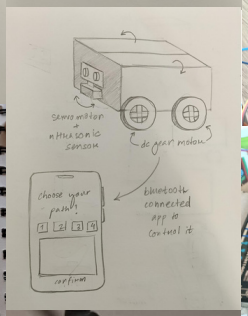
- On the software side:
  - Made the circuit diagrams (all the iterations and changes)
  - Flow of the code
  - writing and debugging all code modules
  - Imports, Pin setup and initializations (rewrote the whole thing)
  - BLE server initialisation
  - Ultrasonic sensor function
  - Motor control functions
  - Obstacle avoidance Logic
  - Debugged predefined path Storage and path execution function
  - Main Loop
  - Light Detection (neopixel setup functions)
  - MIT app inventor blocks

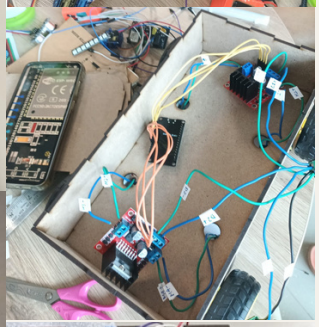
- On the hardware side
  - Assembled the full circuit
  - diagnosing and resolving all electrical issues

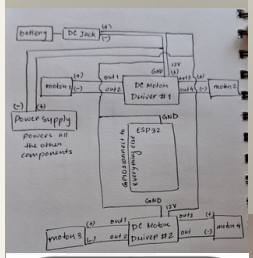
- For fabrication:
- Modelling and laser cutting both the circuit enclosure and the delivery box
- Decorating the outside
- Assembling the majority of the physical structure

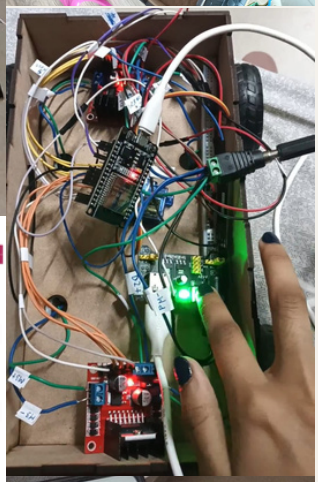


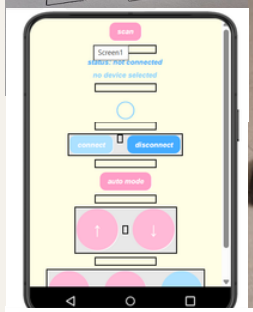


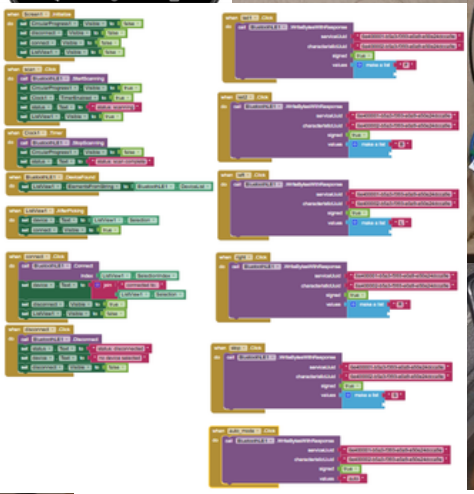


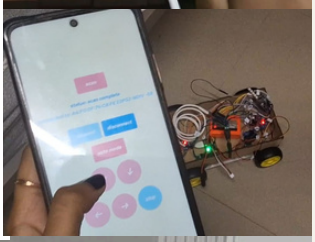


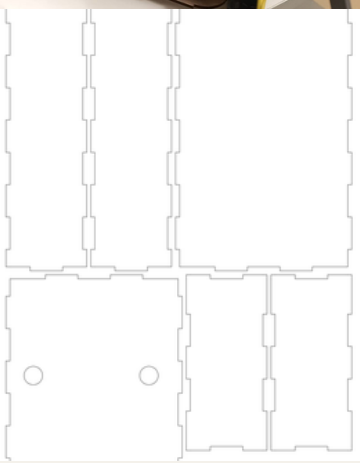












PROJECT TIMELINE & MILESTONES

WEEK	GOALS	DELIVERABLES	STATUS
Week 0	Project ideation, finalise concept and features	Approved project concept	Final idea approved
Week 1	Finalise structure, circuit diagram, features, components, assigning responsibilities	Block diagram, project brief, component list, code flow	Diagrams and brief finalised and approved
Week 2	Begin coding, test individual components, finalise laser cutting files in Rhino	Digital circuit diagram, tested components, Rhino files ready for fabrication, initial draft of the code	Components tested, code in progress, files ready
Final Week	Complete code, laser cut and assemble enclosures, wire the whole circuit, integrate app, test all three modes	Fully functional car, assembled structure, working BLE app, labelled circuit	Tested and functional, debugging completed

WEEKLY PROGRESS LOG

WEEK 1:

- **Planned Tasks:** block diagrams, iterations, sketch block diagrams, define three operating modes, list required components, distribute responsibilities
- **Completed Tasks:** Researched car projects that use ESP32 and L298N through tutorials, watched videos on self-driving delivery robots mechanism, found and adapted reference circuit diagrams for two drivers, finalised the concept as an Indian lorry, defined all three modes and made the first draft of the code, and completed the block diagram, component list, and responsibility matrix.
- **Challenges Faced:** Determining component compatibility and how to power the circuit as the ESP32 operates at 3.3V while L298N and HC-SR04 are nominally 5V and the motors need 6V.
- **Solutions:** Cross referenced multiple tutorials to confirm the connections and power source, took faculty advise and asked what other students with similar concepts were using.
- **Next Steps:** Begin coding individual parts, prepare Rhino files, test components on arrival.

WEEK 2:

- **Planned Tasks:** Write and test all individual codes together, test newly acquired DC motors, finalise laser cutting files in Rhino for both enclosures and box
- **Completed Tasks:** Completed digital circuit diagram with all pin assignments and power rails mapped, tested DC motors individually to confirm direction and speed control, began writing code for motor control functions, BLE server initialisation, and ultrasonic sensor routine, Rhino files for both boxes finalised
- **Challenges Faced:** figuring out the wiring of the DC motors with the other components to ensure that it moves in the correct direction, Rhino files had dimensional and kerfing errors on first pass.
- **Solutions:** Wrote a simple test script to confirm motor rotation before full integration; corrected Rhino files by recalculating material thickness offsets and changing the structure so it doesn't require kerfing.
- **Next Steps:** Laser cut approved files, begin assembling circuit and structure, and begin testing the code and debugging.

FINAL WEEK:

- **Planned Tasks:** Complete and integrate all three operating modes into main loop, laser cut and assemble both enclosures, wire full circuit, test everything with the app
- **Completed Tasks:** Fully functional code with all three modes integrated, both laser-cut enclosures assembled, complete circuit wired and individually labelled, MIT App Inventor app, physical on/off switch installed on exterior panel, testing and debugging completed across all three modes
- **Challenges Faced and Solutions:**
  - Wire management - with two motor drivers, ESP32, ultrasonic sensor, power module, neopixel, servo motor and battery all in one circuit, wires were tangled and difficult to trace. Labelled every wire individually and used the breadboard power rail as a shared bus to reduce crowding.
  - ESP32 not powering from battery - ESP32 was not receiving power due to a missing VIN connection to the supply rail. Connected VIN to the positive rail but still didn't work so later used a USB cable from the power module output for stable regulated power.
  - Auto mode not functioning / controls freezing - the ultrasonic sensor returned None when no obstacle was present, causing an OS error that terminated the main loop. Fixed with a try/except block
  - Left rear wheel only rotating forward - OUT4 on the rear L298N driver had no voltage output. Rewired the left rear motor to the channel on the front driver.
  - Left side wheels slower than right - uneven load from electronics maybe
- **Next Steps:** Record final video, compile all documentation, upload to GitHub